

Unit 1: Cognitive Psychology

Definition & Scope

"Study of how people perceive, learn, remember, and think about information" (Neisser, 1967). Also: "scientific study of the mind." Behaviour is determined by **intentionality, goals, plans** — mental phenomena that affect behaviour.

12 Domains

Cognitive Neuroscience · Perception · Pattern Recognition · Attention · Consciousness · Memory · Knowledge Representation · Imagery · Language · Developmental Psychology · Thinking & Concept Formation · Human & AI

Research Methods (4)

- **Lab experiments:** IV/DV/control vars → probable causality
- **Psychobiological:** post-mortem, brain imaging of deficits, imaging during normal cognition
- **Self-reports/case studies/naturalistic observation:** hypothesis formulation
- **Computer simulations & AI:** model cognitive architecture
- **Cognitive science** = cross-disciplinary (psychology, psychobiology, AI, philosophy, linguistics, anthropology)

History (3 Periods)

- **Early:** Empiricists (experience) vs. Nativists (innate). Aristotle — heart; Plato — brain
- **Renaissance:** *mundus sensibilis* + *mundus intellectualis*. British Empiricists: 3 types of representation
- **Modern:** 19th C: Wundt/Titchener (structure) vs. Brentano (process). James (1890). Behaviourism → Neocognitive Revolution (1950s)

Key pre-cursors: Tolman (1932) **cognitive map** · Bartlett (1932) **schema**
6 forces of revival: failure of behaviourism · communication theory · linguistics · memory research · computer science · Piaget

Key Issues

Nature vs Nurture · Rationalism vs Empiricism · Structures vs Processes · Domain generality vs specificity · Validity vs Ecological validity · Applied vs Basic · Biological vs Behavioural

Computer Metaphor

Cognition = computational (mind = software, brain = hardware). **Neural networks** — act more like humans. **Cognitive neuroscience:** MRI, PET, EEG show structures + processes.

Unit 2: Information Processing

4 Principles (Huiitt, 2000)

- **Limited capacity** — bottlenecks (Broadbent 1975; Case 1978)
- **Control mechanism** — executive function (Atkinson & Shiffrin, 1971)
- **Two-way flow** — bottom-up (sensory) + top-down (memory)
- **Biological preparedness** — infants prefer faces; universal language milestones

Bloom's Taxonomy (1956/1965)

3 domains: **Cognitive** · Psychomotor · Affective. 6 levels: **Knowledge** → **Comprehension** → **Application** → **Analysis** → **Synthesis** → **Evaluation**. Anderson & Krathwohl (2000): reverse top two. Hummel & Huiitt (1994): first 4 confirmed hierarchical.

Sternberg's Info Processing (1988)

Development = skills-based, continuous. 3 components: **Metacomponents** (executive/planning) · **Performance components** (executing) · **Knowledge-acquisition components** (learning). Part of Triarchic Theory. Novice → expert via feedback, self-monitoring, automatisisation.

Unit 3: Memory & Brain (Neuropsychological Basis)

Three Stages (Squire, 1987)

Encoding (acquisition + consolidation) → **Storage** → **Retrieval**

Case H.M. (Scoville & Milner, 1953)

Bilateral medial temporal lobe removal. **Severe anterograde amnesia**; normal STM. Hippocampus critical for forming new long-term memories. **Case R.B.:** lesions restricted to **CA1 pyramidal cells** in hippocampus.

Medial Temporal Lobe Structures

Amygdala · Hippocampus · Entorhinal cortex · Parahippocampal cortex · Perirhinal cortex (Brodman 35, 36). **Amygdala = emotional memory** (Zola et al., 1993) — NOT part of LTM acquisition system.

Imaging Studies

HERA model (Haxby & Ungerleider, 1996): encoding = left hemisphere; retrieval = right; dorsolateral prefrontal cortex.
Schacter et al. (1996): implicit & explicit retrieval = separate brain systems. Occipital cortex (area 19) deactivation during priming.
Procedural memory: motor cortex, putamen, basal ganglia (Grafton et al., 1995).
Semantic retrieval: left prefrontal cortex (Broca's area, Brodmann 44/45/46).

Cellular Bases

- **Hebb's Law (1949):** repeated co-activation strengthens synapses — Hebbian learning
- **Long-Term Potentiation (LTP):** hippocampal circuits → enhanced sensitivity, lasts weeks. Mechanism for long-term retention
- **NMDA receptor:** glutamate = excitatory transmitter. AP5 blocks NMDA → prevents LTP induction, not maintenance. NMDA central to *producing* LTP; non-NMDA to *maintaining* it

Memory Consolidation

Biological changes underlying long-term retention. Hippocampus **coordinates** consolidation but is **NOT the storage repository** — effects in neocortex. Once consolidated, hippocampus no longer required. Retrograde amnesia = temporally limited (1–few years).
Korsakoff's syndrome: dorsomedial thalamus + mammillary bodies. Chronic alcoholism → vitamin deficiency → brain damage.

Unit 4: Memory Models

Atkinson & Shiffrin (1968) — Multi-Store

Sensory Register (½ s vision, 3–5 s hearing; Sperling 1960; Darwin 1972) → **STS/Working Memory** (5 ± 2, Miller 1956; 20–30 s; maintenance vs elaborate rehearsal) → **LTS** (meaning + importance; episodic/semantic, Tulving 1972; declarative/procedural/imagery, Paivio). Memory ≠ Store. Direct SR → LTS pathway possible.

Waugh & Norman (1965)

PM (short-term, displacement, verbatim recall) vs SM (longer-term, rehearsal). 16-digit probe task: **interference > decay**. "Boxes in the head."

Levels of Processing (Craik & Lockhart, 1972)

Shallow sensory → deeper semantic. Deeper = better retention. Maintenance rehearsal (same level) vs elaborate rehearsal (deeper).
3 criticisms (Baddeley, 1978): mundane · untestable · circular
Craik & Tulving (1975): structural < phonological < semantic. **Self-reference effect** (Rogers et al., 1977). **Zinchenko (1962):** LOR.

LTM Organisation

Schemas = interrelated networks in LTM. **Top-down processing** shapes perception (bias, optical illusions). **Chunking** aids STM → LTM transfer.